

Advanced Time Series Analysis and Forecasting

Chapter 9: VaR, ES and backtesting: elicitability, scoring and model risk



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Acronyms used in this material (1/3)

- **ACI**: Adaptive Conformal Inference
- **ADAPT**: Adaptive (CAViaR specification)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **ARMA-GARCH**: ARMA mean with GARCH variance
- **AS**: Asymmetric Slope (CAViaR specification)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- **DM**: Diebold–Mariano (test)
- **DQ**: Dynamic Quantile (test of Engle and Manganelli)
- **EDF**: Empirical Distribution Function
- **EM**: Engle and Manganelli (2004), CAViaR
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **EVT**: Extreme Value Theory
- **EWMA**: Exponentially Weighted Moving Average
- **FHS**: Filtered Historical Simulation
- **FZ**: Fissler–Ziegel (class of scoring functions for VaR and ES)
- **FZ-1F**: One-factor GAS model for VaR and ES (label of Patton, Ziegel and Chen)
- **FZ-2F**: Two-factor GAS model for VaR and ES (label of Patton, Ziegel and Chen)

Acronyms used in this material (2/3)

- **FZ0**: Zero-homogeneous Fissler–Ziegel loss (Patton, Ziegel and Chen)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-EDF**: GARCH with innovations from the empirical distribution function
- **GARCH-FZ**: GARCH model estimated by FZ0 minimisation
- **GARCH-N**: GARCH with Normal innovations
- **GAS**: Generalised Autoregressive Score (model)
- **GAS-1F**: One-factor GAS model for VaR and ES
- **GAS-2F**: Two-factor GAS model for VaR and ES
- **GAS-FZ**: GAS models for VaR and ES driven by the FZ0 loss
- **GCH**: GARCH (short label in tables)
- **GCH-EDF**: ARMA-GARCH with innovations from the empirical distribution function
- **GCH-FZ**: GARCH model estimated by FZ0 minimisation (table label)
- **GCH-N**: ARMA-GARCH with Normal innovations
- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **GM**: General Motors (stock)
- **GPL**: Generalised Piecewise Linear (scoring function)
- **GW**: Giacomini–White (test of conditional predictive ability)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HC0**: Heteroskedasticity-Consistent covariance of White (type 0)
- **HS**: Historical Simulation
- **IBM**: International Business Machines (stock)

Acronyms used in this material (3/3)

- **IG**: Indirect GARCH (CAViaR specification)
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio (test)
- **MA**: Moving Average
- **MAR33**: Basel Framework, chapter MAR33 (internal models, capital requirements)
- **MCS**: Model Confidence Set
- **MFM**: Modelling Financial Markets
- **MLE**: Maximum Likelihood Estimation
- **MSE**: Mean Squared Error
- **PIT**: Probability Integral Transform
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **PZC**: Patton, Ziegel and Chen (2019)
- **QLIKE**: Quasi-LIKElihood loss
- **QML**: Quasi-Maximum Likelihood
- **RQ**: Regression Quantile (criterion of Engle and Manganelli)
- **RW**: Random Walk
- **RW-125**: Rolling Window of 125 days (historical simulation)
- **RW-250**: Rolling Window of 250 days (historical simulation)
- **RW-500**: Rolling Window of 500 days (historical simulation)
- **SAV**: Symmetric Absolute Value (CAViaR specification)
- **TSA**: Time Series Analysis
- **VAR**: Vector AutoRegression
- **VIX**: CBOE Volatility IndeX
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and route

- **Question**: a VaR or an ES forecast is a point forecast of a tail functional; how do we estimate dynamic models for it, test whether it is calibrated and decide which forecast is better?
  - the tail is observed rarely: a forecast must be judged with a loss that is consistent for its target, on few informative days
- **Route** of the chapter
  - risk measures as functionals; elicitability, the Fissler–Ziegel class, FZ0 and Murphy diagrams
  - quantile regression and CAViaR; semiparametric (VaR, ES) models and joint regression
  - backtesting as conditional calibration; estimation risk in backtests; comparison with the MCS in stress periods
  - the horizon (square-root-of-time), model risk, extremes under dependence, conformal calibration
- We build on Chapter 1 (scoring rules, elicitability, DM, MCS), Chapter 0 (HAC, bootstrap), Chapter 8 (volatility) and TSA, Chapter 5 (GARCH); Seminar 9 comes before this lecture

Learning outcomes

- Characterise the scoring functions consistent for quantiles and for the pair (VaR, ES), and prove that ES alone is not elicitable
- Estimate CAViaR models with the Engle–Manganelli procedure, compute their standard errors and run the DQ test
- Estimate semiparametric dynamic (VaR, ES) models by FZ0 minimisation and replicate the out-of-sample comparison of Patton, Ziegel and Chen
- Test conditional calibration with identification functions, account for estimation risk, and compare risk models with DM tests, Murphy diagrams and the MCS
- Quantify horizon, model and estimation risk, and correct a miscalibrated VaR by adaptive conformal inference

Reading, data and tools

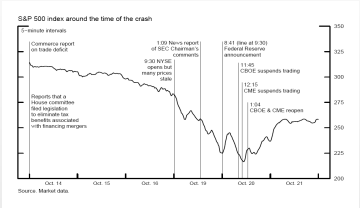
- Theory: Gneiting (2011); Fissler and Ziegel (2016); Ehm et al. (2016); Nolde and Ziegel (2017)
  - Models: Koenker and Bassett (1978); Engle and Manganelli (2004); Patton, Ziegel and Chen (2019); Dimitriadis and Bayer (2019)
  - Survey of forecasting practice: Petropoulos et al. (2022)
- Python Quantlets of this chapter: Quantlets/Ch_09
  - CAViaR, the GAS-FZ models, the joint regression, the backtests, DM and the MCS written in `numpy/scipy`; `numba` only speeds up the recursions
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source, sample	In sample / out of sample
S&P 500, DAX	EODHD daily closes, January 1990 – September 2026	1990–1999 / 2000–2026
BET	EODHD, official closes, 2000–2026	2000–2009 / 2010–2026
EUR/RON	BNR reference rate, July 2005 – September 2026	to June 2015 / from July 2015
Bitcoin	EODHD, 7 days a week, September 2014 – September 2026	2014–2019 / 2020–2026
VIX	EODHD daily closes	regressor of the joint (VaR, ES) regression

- Daily log returns in %; ten in-sample years, as in Patton, Ziegel and Chen (2019), five for Bitcoin
- Four stress periods, fixed before looking at the losses
  - September 2008 – March 2009; 19 February – 30 June 2020; the calendar year 2022; March – June 2025

A short history of VaR, ES and backtesting



The S&P 500 around 19 October 1987

Figure: Mark Carlson, Federal Reserve Board (2006); public domain; Wikimedia Commons

- 1987: a one-day fall of about 20% in US equities shows that the tail, not the variance, decides solvency
- 1996: the Basel Committee adopts VaR for market risk and a traffic-light backtest of its hits Basel Committee (1996)
- 1999–2002: VaR is not coherent; ES is Artzner et al. (1999), Acerbi and Tasche (2002)
- 2011–2016: ES is not elicitable Gneiting (2011), but (VaR, ES) is jointly elicitable Fissler and Ziegel (2016)
- FRTB: capital on ES 2.5%, backtests on VaR 1% and VaR 2.5% Basel Committee (MAR33)

Basel: bank risk forecasts and supervisory backtests



Bank for International Settlements, Basel, 2018

Photo: Michal Pleskowicz (2018); CC BY-SA 4.0; Wikimedia Commons

- A bank issues a daily forecast of its own tail; the supervisor checks it after the fact
- Two questions that statistics separates: is the forecast **calibrated** (backtest) and is it **better** than another (comparison)?
- Elicitability decides which forecasts can be compared at all Nolde and Ziegel (2017)
- The incentive matters: a scoring rule rewards honest reporting only if it is strictly consistent for the target

Known from TSA and MFM, and new here

- Known: VaR and ES by historical simulation, parametric, filtered HS and EVT; Kupiec and Christoffersen tests; the Basel traffic light (TSA, Chapter 5; MFM, Chapters 7–8)
- New: the forecasting view
  - which losses are consistent for a quantile and for (VaR, ES); what elicitability buys and what it does not
  - M-estimation of dynamic quantile and ES models: identification, asymptotics, non-smooth objectives
  - backtests as moment tests of conditional calibration, with estimation risk; comparisons that survive multiple testing
- Replications: Engle and Manganelli (2004) and Patton, Ziegel and Chen (2019), with the specifications of the papers, on our data

VaR and ES as functionals (1/2): definitions

- VaR is minus a quantile of the return distribution

$$q_\alpha(F) = \inf\{y : F(y) \geq \alpha\}, \qquad \mathrm{VaR}_\alpha(F) = -q_\alpha(F)$$

  - Y: the return, with distribution function F; α: the level, i.e. the target hit rate (VaR 1%: $\alpha = 0.01$)
  - $q_\alpha(F)$: the smallest y with $F(y) \geq \alpha$; a correct VaR is exceeded with probability $\Pr(Y < -\mathrm{VaR}_\alpha) = \alpha$

- ES is minus the average of the quantiles below level α Acerbi and Tasche (2002)

$$\mathrm{ES}_\alpha(F) = -\frac{1}{\alpha}\int_0^\alpha q_u(F)\,du$$

  - for continuous F: $\mathrm{ES}_\alpha = -\mathbb{E}[Y \mid Y \leq q_\alpha]$, minus the mean return on the worst α share of days
  - ES is coherent (subadditive) Artzner et al. (1999); VaR is not; ES needs $\mathbb{E}|Y| < \infty$

VaR and ES as functionals (2/2): conventions

- In formulas we work on the return scale with the pair $(v, e) = (q_\alpha, -\mathrm{ES}_\alpha)$
  - v: the α-quantile of returns; e: the mean return beyond it; for the usual levels both are negative and $e \leq v < 0$
- Both are **functionals** $\mathrm{T}(F)$: maps from a distribution to a number
  - a forecast of them is a point forecast of the predictive distribution, as in Chapter 1

Conditional risk forecasts: three routes (1/2)

- Target: the functional of the conditional distribution

$$(v_t, e_t) = \mathrm{T}(F_{t|t-1})$$

  - $F_{t|t-1}$: distribution of the return Y_t given $\mathcal{F}_{t-1}$, the information up to day $t-1$
- **Location–scale**: the return is a mean plus a volatility times an i.i.d. shock

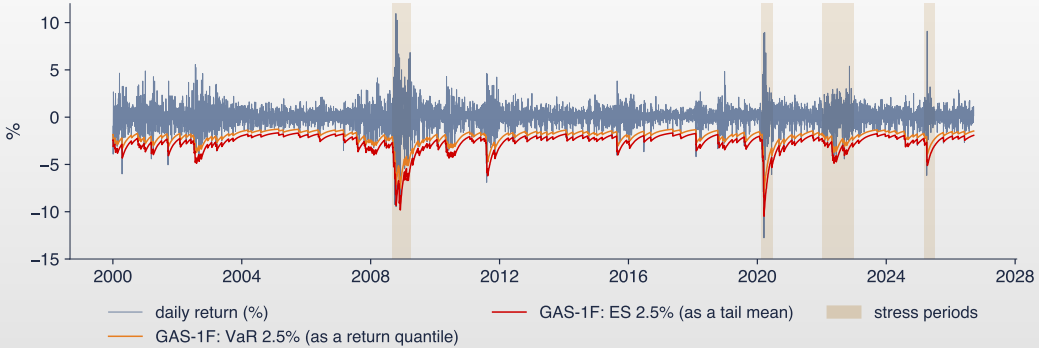
$$Y_t = \mu_t + \sigma_t \eta_t \;\Rightarrow\; v_t = \mu_t + \sigma_t\, q_\alpha(\eta), \qquad e_t = \mu_t + \sigma_t\, \mathbb{E}[\eta \mid \eta \le q_\alpha(\eta)]$$

  - μ_t, σ_t: conditional mean and volatility (ARMA, GARCH); $q_\alpha(\eta)$: quantile of the shock distribution
  - GARCH with Normal, skewed-t Hansen (1994) or empirical (EDF) innovations; the ratio e_t/v_t is constant over time when $\mu_t = 0$

Conditional risk forecasts: three routes (2/2)

- **Direct quantile dynamics** Engle and Manganelli (2004): $v_t = v(\mathcal{F}_{t-1}; \beta)$
  - $v(\cdot; \beta)$: a recursion with parameters β, estimated by the quantile (pinball) loss, without a distribution
- **Joint semiparametric**: $(v_t, e_t) = (v, e)(\mathcal{F}_{t-1}; \theta)$
  - two recursions with parameters θ, estimated by a Fissler–Ziegel loss Patton, Ziegel and Chen (2019), Dimitriadis and Bayer (2019)
- Every route produces a point forecast of a functional; the scoring function of the next section is what makes them comparable

Thirty-six years of tail forecasts



- S&P 500 daily returns; the GAS-1F model of Patton, Ziegel and Chen, parameters estimated once on 1990–1999 and then only filtered; shaded: the four stress periods

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Interpreting the long-run forecasts

- The ES 2.5% forecast ranges from about -2.44% (median) to -10.5% on 17 March 2020: tail risk moves as much as volatility
- The forecasts react only on hit days and decay smoothly otherwise: the score-driven update of the GAS model
- Parameters fixed in 1999 still track 2008, 2020 and 2025: the dynamics generalise, the level does not need re-estimation
- Whether this is good enough is a testing question (Sections 5–6), not a visual one

Recap: Risk measures as functionals

- VaR is a quantile, ES an integral of quantiles; both are functionals of the predictive distribution
- Three routes: location–scale, direct quantile dynamics, joint semiparametric models
- The convention: VaR 1%, ES 2.5%, α is the target hit rate

Consistency, elicitability, identification (1/2)

- A scoring function $S(x, y)$ assigns a loss to the forecast x when y is observed; it is **consistent** for T on a class $\mathcal{F}$ if Gneiting (2011)

$$\mathbb{E}_F S(\mathrm{T}(F), Y) \leq \mathbb{E}_F S(x, Y) \quad \text{for all } x \text{ and } F \in \mathcal{F}$$

  - $\mathbb{E}_F$: expectation when $Y \sim F$; the true functional value has the smallest expected loss
  - **strictly** consistent: equality forces $x = \mathrm{T}(F)$; T is **elicitable** if a strictly consistent S exists (Chapter 1)

Consistency, elicitability, identification (2/2)

- **Identification function**: a function $V(x, y)$ whose mean is zero exactly at the true value

$$\mathbb{E}_F V(x, Y) = 0 \iff x = \mathrm{T}(F)$$

  - quantile: $V(x, y) = 1\{y \leq x\} - \alpha$, with $1\{\cdot\}$ the indicator (1 if true, 0 otherwise); mean: $V(x, y) = x - y$
- Osband's principle: the slope of the expected score is a positive multiple of the expected identification function Fissler and Ziegel (2016)

$$\partial_x \mathbb{E}_F S(x, Y) = h(x)\, \mathbb{E}_F V(x, Y), \qquad h > 0$$

  - scores are integrated identification functions
- Identification functions give **backtests** (is the forecast calibrated?); scoring functions give **comparisons** (which forecast is better?) Nolde and Ziegel (2017)

Quantiles: all consistent scores

- **Theorem** Gneiting (2011): under mild conditions, S is consistent for q_α if and only if it is generalised piecewise linear (GPL)

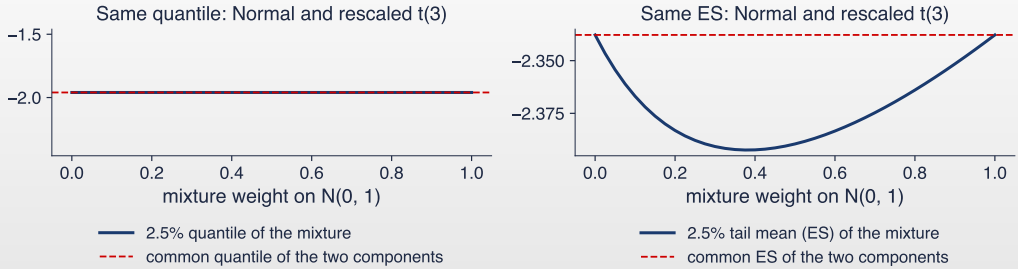
$$S(x, y) = (1\{y \le x\} - \alpha)\big(G(x) - G(y)\big), \qquad G \text{ non-decreasing}$$

  - $G(x) = x$: the pinball loss; G strictly increasing gives strict consistency; $G(x) = \ln x$ for positive variables gives a scale-free (zero-homogeneous) score
  - why: $\partial_x \mathbb{E}_F S(x, Y) = G'(x)\,(F(x) - \alpha)$, negative below q_α, positive above (Appendix)
- Different G give the same optimal forecast but can rank two misspecified forecasts differently: the choice of G is a modelling choice
- Expectiles (asymmetric squared loss) are the only elicitable coherent risk measures Ziegel (2016); Taylor (2008) uses them for VaR and ES

ES is not elicitable

- **Necessary condition** Gneiting (2011): if T is elicitable, its level sets are convex
  - level set: $\{F : \mathrm{T}(F) = t\}$, all distributions with the same functional value t; convex: closed under mixtures $\lambda F_0 + (1-\lambda)F_1$, $\lambda \in [0,1]$
  - proof: if $\mathbb{E}_{F_0} S(t, Y) \leq \mathbb{E}_{F_0} S(x, Y)$ and the same for F_1, the inequality holds for the mixture, because expectations are linear in F
- Quantiles pass: if $F_0(t) = F_1(t) = \alpha$, then every mixture has $F_\lambda(t) = \alpha$
- ES fails: the mixture changes the quantile, so it changes which part of each component enters the tail mean Weber (2006)
  - consequence: no loss function can rank ES forecasts alone; average losses of ES forecasts are not meaningful by themselves
- Variance fails for the same reason; (mean, second moment) and (VaR, ES) are elicitable as pairs (Fissler and Ziegel (2016), "higher order elicitability")

Level sets under mixing



- Left: $N(0,1)$ and a $t(3)$ rescaled to the same 2.5% quantile; right: $N(0,1)$ and a $t(3)$ rescaled to the same ES 2.5%; curves: the functional of the mixture

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Interpreting the level sets

- The quantile of every mixture stays at -1.96: the level set of $q_{0.025}$ is convex
- Both components have tail mean -2.338; the mixture moves away by up to 0.055 (at weight 0.38): the level set of ES is not convex
- The gap is small in numbers but decisive in logic: one counterexample rules out every strictly consistent loss for ES alone
- The way out is to forecast the quantile together with ES: the pair has a convex level set

Joint elicitability: the Fissler–Ziegel class (1/2)

- **Theorem** Fissler and Ziegel (2016), in the form of Patton, Ziegel and Chen (2019), eq. (4): for $e \leq v$, the loss

$$S(v, e, y) = (1\{y \leq v\} - \alpha)\big(G_1(v) - G_1(y)\big) + G_2(e)\Big(v - e + \frac{1}{\alpha} 1\{y \leq v\}(y - v)\Big) - \mathcal{G}_2(e)$$

  - is consistent for $(q_\alpha, -\mathrm{ES}_\alpha)$ if G_1 is non-decreasing and $G_2 = \mathcal{G}_2'$ is positive and increasing; strictly so under mild conditions
- Reading the two parts
  - first term: a GPL quantile score for v (previous slides); G_1: any non-decreasing function
  - second part: a Bregman-type score for e, evaluated against the tail mean implied by v; $\mathcal{G}_2$: an antiderivative of G_2

Joint elicitability: the Fissler–Ziegel class (2/2)

- Identification function of the pair: one component for the quantile, one for the tail mean

$$V_1 = 1\{y \le v\} - \alpha, \qquad V_2 = e - v + \frac{1}{\alpha} 1\{y \le v\}(v - y)$$

  - $\mathbb{E}V_1 = 0$: v is the α-quantile; $\mathbb{E}V_2 = 0$ at the true quantile is the Acerbi–Tasche formula
  - ES is the tail mean only together with the right quantile
- Consequence Fissler, Ziegel and Gneiting (2016): ES can be **compared** through the pair; traditional ES backtests still need more than ES (Section 5)

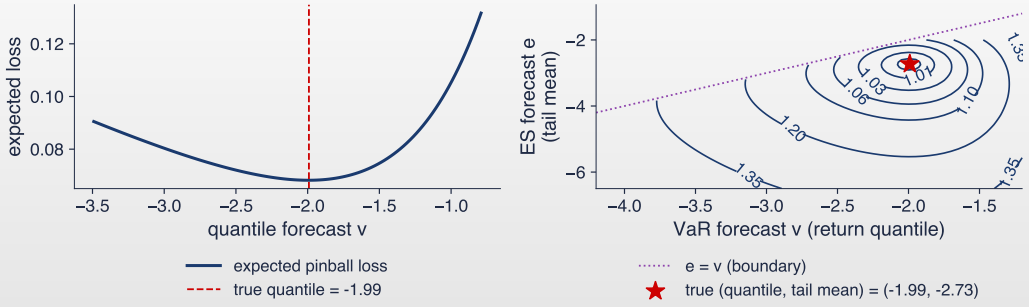
FZ0: the zero-homogeneous member

- Patton, Ziegel and Chen (2019), Proposition 1: with $v, e < 0$, loss **differences** are invariant to rescaling Y if and only if $G_1 = 0$ and $G_2(e) = -1/e$

$$L_{\mathrm{FZ0}}(y, v, e; \alpha) = -\frac{1}{\alpha e} 1\{y \leq v\}(v - y) + \frac{v}{e} + \ln(-e) - 1$$

  - first term: positive only on hit days ($y \leq v$), proportional to the size of the hit; $v/e \in (0, 1]$; $\ln(-e)$ penalises an exaggerated ES
  - the VaR part resembles the pinball loss; the ES part resembles QLIKE (Chapter 8)
- Why zero homogeneity matters for time series
  - with a non-homogeneous score, volatile days dominate the average loss and the DM test (heteroskedastic loss differences)
  - FZ0 ranks forecasts in % and in basis points identically; its value can be negative (EUR/RON)
  - proof of consistency and of homogeneity: Appendix
- Iso-expected-loss contours are convex under mild conditions, which helps numerical minimisation Patton, Ziegel and Chen (2019)

Expected pinball and FZ0 losses



⊡ Student $t(5)$ with unit variance, $\alpha = 0.025$, expectations by simulation (4×10^5 draws): left, expected pinball loss; right, contours of the expected FZ0 loss

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Interpreting the expected losses

- The pinball curve is minimal at -1.98, the true quantile -1.99: consistency in action
- The FZ0 surface is minimal at $(-1.99, -2.73)$, minimum 1.004; contours are convex and elongated
- Both surfaces are flat near the optimum: large forecast differences cost little expected loss, so tests need many tail days
- Forecasts with $e > v$ are inadmissible: estimation must enforce the ordering

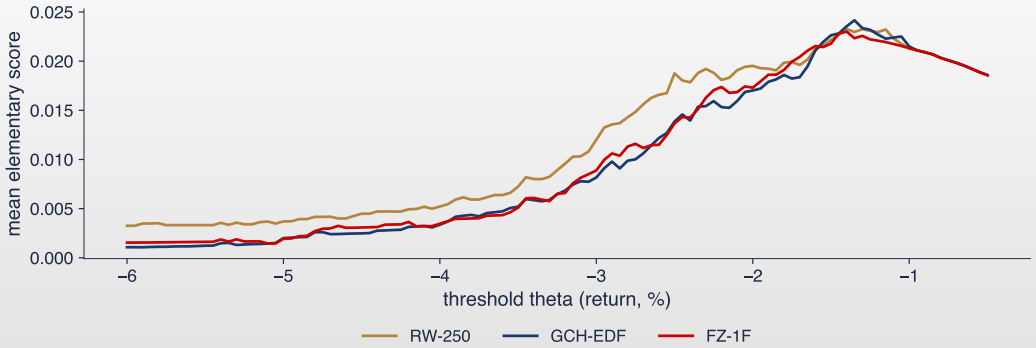
Murphy diagrams: ranking for all consistent scores

- Ehm et al. (2016): every GPL quantile score is a mixture of **elementary scores**

$$S_\theta(x,y) = (1\{y < x\} - \alpha)\big(1\{\theta < x\} - 1\{\theta < y\}\big), \qquad S(x,y) = \int S_\theta(x,y)\, dH(\theta)$$

  - θ: a threshold; S_θ: the loss of a binary decision "is the return below θ?"; H: a non-decreasing weight over thresholds, corresponding to G
- **Murphy diagram**: the average $\bar{S}_\theta$ of each forecast, plotted against θ
  - forecast A dominates B for every consistent score if and only if its curve is below for all θ
  - crossing curves: the ranking depends on G, i.e. on which thresholds the user cares about
- A check on the pinball ranking that costs one line of code

A Murphy diagram for VaR 2.5%



- S&P 500, out of sample 2000–2016 (PZC design); mean elementary scores of three VaR 2.5% forecasts against the threshold θ

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Interpreting the Murphy diagram

- GARCH-EDF lies below RW-250 at 91% of the thresholds: the dynamic forecast dominates almost uniformly
- GAS-1F and GARCH-EDF cross: GAS-1F is better at only 38% of the thresholds; average pinball 7.66 and 7.45 ($\times 10^{-2}$)
- The two good models differ in deep thresholds (below -4%), where data are scarce
- A claim that "model A is better" should say for which scores; the diagram answers it

Recap: Elicitability

- Scores compare, identification functions test; both come from the same theory
- Quantiles: GPL scores; ES: not alone; (VaR, ES): the Fissler–Ziegel class
- FZ0 is scale-free; Murphy diagrams check rankings for all consistent scores

Quantile regression as M-estimation (1/2)



Roger Koenker, Oberwolfach, 2012

Photo: Ivonne Vetter, MFO (2012); CC BY-SA 2.0 de; Wikimedia Commons

- Koenker and Bassett (1978): the α-quantile of y_t given x_t is modelled as $x_t'\beta$ and estimated by minimising the pinball loss

$$\hat{\beta}(\alpha) = \arg\min_{\beta} \sum_t \rho_\alpha(y_t - x_t'\beta), \qquad \rho_\alpha(u) = u\,(\alpha - 1\{u < 0\})$$

  - x_t: vector of regressors; ρ_α: the check (pinball) function; the problem is a linear program
  - ρ_α has slope α for positive residuals and $\alpha - 1$ for negative ones
  - first-order condition: $\sum_t x_t(1\{y_t \leq x_t'\hat{\beta}\} - \alpha) \approx 0$, the identification function

Quantile regression as M-estimation (2/2)

- Asymptotic distribution Koenker (2005): a sandwich in which the density at the quantile appears

$$\sqrt{T}(\hat{\beta} - \beta) \to N(0, \alpha(1-\alpha)D^{-1}\Omega D^{-1}), \qquad \Omega = \mathbb{E}[x_t x_t'], \qquad D = \mathbb{E}[f_t(q_t) x_t x_t']$$

  - $f_t(q_t)$: conditional density of y_t at its true quantile q_t; its inverse is called the "sparsity"
  - few observations near the quantile (small f_t) mean an imprecise $\hat{\beta}$; f_t is estimated with a kernel and a bandwidth
- Time series: quantile autoregression Koenker and Xiao (2006) lets AR coefficients vary across quantiles; CAViaR makes the quantile itself autoregressive

CAViaR: the four specifications



Robert Engle, 2022

Photo: Jon Demske (2022); CC BY-SA 4.0; Wikimedia Commons

- Engle and Manganelli (2004), written for $\mathrm{VaR}_t = -q_t(\alpha) > 0$; y_{t-1}: yesterday's return; β_j: parameters:
- **SAV** (symmetric absolute value): $\mathrm{VaR}_t = \beta_1 + \beta_2 \mathrm{VaR}_{t-1} + \beta_3 |y_{t-1}|$
  - β_2: the persistence of the tail; β_3: the reaction to the size of yesterday's return
- **AS** (asymmetric slope): $\mathrm{VaR}_t = \beta_1 + \beta_2 \mathrm{VaR}_{t-1} + \beta_3 (y_{t-1})^+ + \beta_4 (y_{t-1})^-$
  - $(y)^+ = \max(y, 0)$, $(y)^- = -\min(y, 0)$: gains and losses enter separately; $\beta_4 > \beta_3$ is a leverage effect
- **IG** (indirect GARCH): $\mathrm{VaR}_t = (\beta_1 + \beta_2 \mathrm{VaR}_{t-1}^2 + \beta_3 y_{t-1}^2)^{1/2}$, a GARCH(1,1) with i.i.d. innovations
- **Adaptive**: $\mathrm{VaR}_t = \mathrm{VaR}_{t-1} + \beta_1 \left([1 + e^{G(y_{t-1} + \mathrm{VaR}_{t-1})}]^{-1} - \alpha \right)$
  - the bracket is a smooth hit indicator ($G = 10$): VaR rises after a hit and falls slowly otherwise; no distribution is assumed in any of the four

Estimation and inference (1/2): the criterion

- Regression-quantile criterion: the average pinball loss of the implied quantile path

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{T} \sum_{t} \rho_{\alpha}\left(y_t + \mathrm{VaR}_t(\beta)\right)$$

  - $\mathrm{VaR}_t(\beta)$: the path generated by the recursion with parameters β; the criterion is non-differentiable and non-convex in β
- EM procedure (empirical section)
  - VaR_1: the empirical quantile of the first 300 days; 10^4 random parameter vectors, the 10 best refined by alternating simplex and quasi-Newton steps

Estimation and inference (2/2): the covariance

- Consistency and asymptotic normality, with a sandwich covariance

$$\sqrt{T}(\hat{\beta} - \beta) \to N\big(0, \alpha(1-\alpha)D^{-1}AD^{-1}\big), \qquad A = \mathbb{E}[\nabla q_t \nabla q_t'], \qquad D = \mathbb{E}[f_t(q_t)\nabla q_t \nabla q_t']$$

  - $\nabla q_t = \partial q_t(\beta)/\partial\beta$: gradient of the quantile path, which follows its own recursion (or numerical differentiation of the path)
  - $f_t(q_t)$: conditional density of y_t at the quantile, as in quantile regression
- Estimator of D: count the residuals within $\hat{c}$ of zero

$$\hat{D} = \frac{1}{2T\hat{c}} \sum_t 1\{|y_t - \hat{q}_t| < \hat{c}\}\, \nabla\hat{q}_t \nabla\hat{q}_t'$$

  - $\hat{c}$: bandwidth, from the Hall–Sheather rule on the residual scale Koenker (2005); the dynamic model is a recursive M-estimator

The dynamic quantile (DQ) test

- The hit sequence is a martingale difference under correct specification

$$\mathrm{Hit}_t = 1\{y_t < q_t\} - \alpha, \qquad \mathbb{E}[\mathrm{Hit}_t \mid \mathcal{F}_{t-1}] = 0$$

  - regress Hit_t on $X_t = (1, \mathrm{Hit}_{t-1}, \ldots, \mathrm{Hit}_{t-4}, q_t)$, all in $\mathcal{F}_{t-1}$ Engle and Manganelli (2004)
- Out-of-sample statistic: a Wald test that all regression coefficients are zero

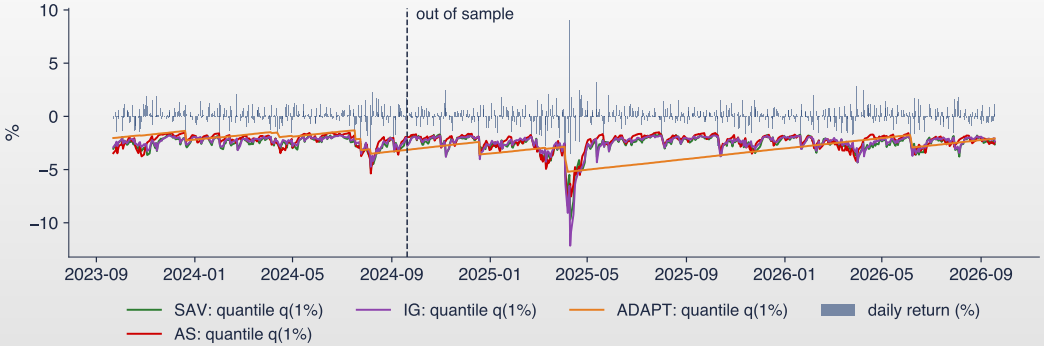
$$\mathrm{DQ} = \frac{\mathrm{Hit}' X (X'X)^{-1} X' \mathrm{Hit}}{\alpha(1-\alpha)} \to \chi^2_6$$

  - Hit: vector of hits; X: matrix with rows X_t; 6 = number of regressors; a large DQ rejects calibration
  - in sample the estimated β makes the hits orthogonal to the gradients: the covariance needs the correction derived by EM
- Kupiec (constant only) and Christoffersen (the first lag of the hits) are special cases; the VaR regressor adds power against level-dependent misspecification
- With $\alpha = 1\%$ and a few hundred days the χ^2 approximation is poor (Section 5): report simulated or exact p-values when possible

Case study: Engle and Manganelli (2004) on today's data

- The design of the empirical section of EM: 3,392 daily returns, the first 2,892 for estimation, the last 500 out of sample; VaR 1% and 5%
  - EM used 1986–1999 (GM, IBM, S&P 500); we use the S&P 500 from 25 March 2013 to 18 September 2026, out of sample from 20 September 2024
- Same specifications, starting values and search; the April 2025 tariff shock falls in the out-of-sample period
- Questions: which specification fits the 1% tail, are the forecasts calibrated out of sample, and is the response to bad news asymmetric?

CAViaR VaR 1% of the S&P 500



⊡ The last year of the estimation sample and the 500 out-of-sample days; lines: the 1% return quantile $q_t = -\mathrm{VaR}_t$ of the four specifications

ATS_ch9_caviar

Estimates, standard errors and DQ tests

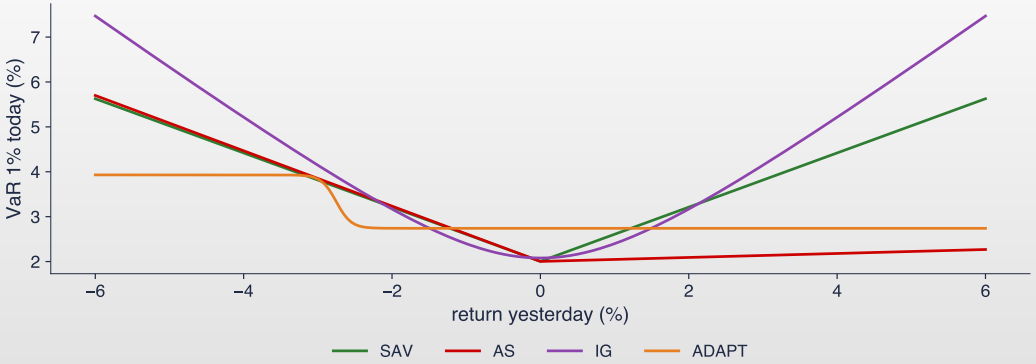
Model	β_1	β_2	β_3	β_4	RQ$\times 10^2$	hits in sample (%)	hits out of sample (%)	DQ out of sample (p)
SAV	0.359 (0.217)	0.702 (0.118)	0.604 (0.243)	–	3.305	1.0	1.4	14.4 (0.025)
AS	0.264 (0.137)	0.805 (0.080)	0.044 (0.185)	0.616 (0.149)	3.220	1.0	1.2	15.8 (0.015)
IG	0.942 (0.467)	0.674 (0.105)	1.429 (0.702)	–	3.267	1.0	1.2	2.3 (0.891)
ADAPT	1.190 (0.069)	–	–	–	4.044	0.9	0.8	27.5 ($<$0.001)

- Standard errors from the asymptotic covariance of EM; RQ: the minimised in-sample criterion; DQ out of sample with four lagged hits and the VaR, χ^2_6
- Out-of-sample average pinball loss ($\times 10^2$): SAV 3.87, AS 3.66, IG 3.66, adaptive 4.42

Interpreting the CAViaR estimates

- All three autoregressive models fit the in-sample rate (about 1%) by construction: the quantile loss calibrates the level
- AS: a negative return moves VaR about 14 times more than a positive one of the same size; β_3 is not distinguishable from zero
- Out of sample IG passes DQ ($p = 0.891$); SAV ($p = 0.025$) and AS ($p = 0.015$) are rejected at 5%, the adaptive model clearly ($p < 0.001$)
- Five expected hits in 500 days: rejections come from clustered hits in April 2025, not from the count; such evidence is fragile
- The adaptive model reacts only to hits, like a slow thermostat: after the April 2025 cluster it stays too high for months

News impact curves



- VaR_t as a function of $y_{t-\mathbf{1}}$, with $\mathrm{VaR}_{t-\mathbf{1}}$ at its in-sample median (2.16% for AS); estimated parameters of the table

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Interpreting the news impact curves

- SAV and IG are symmetric by design: a 5% rally raises the 1% VaR as much as a 5% fall
- AS is almost flat for gains and steep for losses: the leverage effect estimated directly in the tail
- The adaptive curve is a step: only the sign of the hit matters, not its size
- The shape of the curve is a testable restriction: AS nests SAV ($\beta_3 = \beta_4$), a Wald test with the asymptotic covariance of EM

Recap: CAViaR

- Quantile regression is M-estimation with the pinball loss; its covariance needs the density at the quantile
- CAViaR makes the quantile autoregressive; estimation needs many starts
- DQ tests hits against information; the asymmetric model fits equity tails

M-estimation with the FZ0 loss

- A dynamic model for the pair, estimated by minimising the average FZ0 loss Patton, Ziegel and Chen (2019)

$$(v_t, e_t) = (v, e)(\mathcal{F}_{t-1}; \theta), \qquad \hat{\theta} = \arg\min_{\theta} \frac{1}{T} \sum_t L_{\mathrm{FZ0}}\big(y_t, v_t(\theta), e_t(\theta); \alpha\big)$$

  - θ: parameters of the two recursions; consistency follows from the strict consistency of FZ0 and the identification of θ
  - asymptotic normality as for CAViaR, with a sandwich covariance
- Why not maximum likelihood?
  - MLE needs the whole distribution; a wrong body or right tail distorts the left tail estimates
  - FZ estimation targets exactly the two tail functionals: less efficient if the model is right, robust if it is not (PZC, Tables 3–4)
- Related: the asymmetric Laplace quasi-likelihood Taylor (2019); ES regressions Dimitriadis and Bayer (2019); score-driven (GAS) dynamics Creal, Koopman and Lucas (2013)

The two-factor GAS model

- Patton, Ziegel and Chen (2019), eq. (9)–(16): the pair follows a VAR(1) driven by the scaled score of FZ0 (generalised autoregressive score, GAS)

$$\begin{pmatrix} v_{t+1} \\ e_{t+1} \end{pmatrix} = w + B \begin{pmatrix} v_t \\ e_t \end{pmatrix} + A\lambda_t$$

  - w: intercept vector; B: diagonal 2×2 persistence matrix; A: 2×2 matrix of reactions to the forcing variables $\lambda_t = (\lambda_{v,t}, \lambda_{e,t})'$

- Forcing variables from the FZ0 score

$$\lambda_{v,t} = -v_t\big(1\{y_t \le v_t\} - \alpha\big), \qquad \lambda_{e,t} = \frac{1}{\alpha} 1\{y_t \le v_t\}\, y_t - e_t$$

  - both are identification functions: zero conditional mean under correct specification
  - the Hessian is scaled with $f_t(v_t) \approx k_\alpha / v_t$, k_α a constant (exact for location–scale returns), so no density is estimated
  - eight parameters; on non-hit days $\lambda_{e,t} = -e_t$ and both risk measures decay deterministically towards their means

One-factor, GARCH-FZ and Hybrid models

- **GAS-1F**, eq. (20): one latent log-scale κ_t multiplies both risk measures

$$v_t = a\, e^{\kappa_t}, \quad e_t = b\, e^{\kappa_t}, \quad \kappa_t = \beta \kappa_{t-1} + \gamma \frac{-1}{e_{t-1}} \Big(\frac{1}{\alpha} 1\{y_{t-1} \le v_{t-1}\} y_{t-1} - e_{t-1} \Big)$$

  - $b < a < 0$: VaR and ES at $\kappa_t = 0$; β: persistence of the log-scale; γ: reaction to the ES forcing variable
  - the intercept ω is not identified together with (a, b) and is fixed at 0
- **GARCH-FZ**, eq. (25)–(26): $\kappa_t^2 = 1 + \beta \kappa_{t-1}^2 + \gamma y_{t-1}^2$, $(v_t, e_t) = (a, b)\kappa_t$: a GARCH whose scale κ_t is fitted to the tail
- **Hybrid**, eq. (27): GAS-1F plus $\delta \ln |y_{t-1}|$ in the recursion of κ_t
  - reacts every day through δ, more strongly on hit days
- In all three the ratio $e_t / v_t = b/a$ is constant: the shape of the tail is fixed, only its scale moves

Estimating non-smooth recursive models

- The FZ0 objective is discontinuous in θ (through $1\{y_t \leq v_t(\theta)\}$) and the recursion propagates every jump
- Appendix C of the paper Patton, Ziegel and Chen (2019): replace the indicator by a logistic approximation in the loss and in the forcing variable

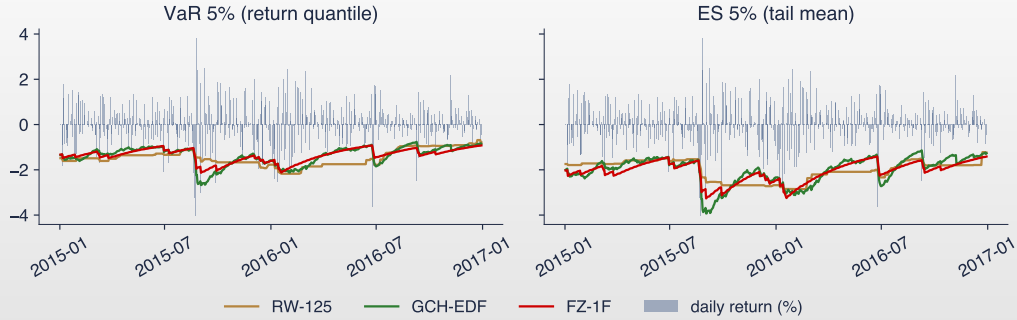
$$\Gamma(y, v; \tau) = \frac{1}{1 + e^{\tau(y-v)}}$$

  - $\tau > 0$: smoothing parameter; as $\tau \to \infty$, $\Gamma \to 1\{y \leq v\}$
  - quasi-Newton with $\tau = 5$, then $\tau = 20$, then the exact loss with the simplex method, each step starting from the previous one
- Our starting values: random dynamic parameters with (a, b) matched to the in-sample VaR and ES; the four best are refined
- Constraints enforced by penalty: $e_t \leq v_t$, $e_t < 0$, stationarity of the recursion

Case study: Patton, Ziegel and Chen (2019), Section 5

- Data: daily S&P 500 returns, January 1990 – December 2016; estimation on the first ten years, parameters kept fixed for 2000–2016 ($T = 4\,275$)
- Ten models, as in the paper
  - rolling windows of 125, 250 and 500 days (RW); ARMA (order by BIC) – GARCH(1,1) with Normal, skewed-t or empirical innovations
  - GAS-2F, GAS-1F, GARCH-FZ and Hybrid estimated by FZ0 minimisation
- Outputs replicated: average out-of-sample FZ0 losses (Table 8, $\alpha = 5\%$; Table S5, $\alpha = 2.5\%$), DM statistics (Table 9), goodness-of-fit tests (eq. 41)
- Our data: EODHD closes of the same index; differences of a few thousandths are expected from data revisions and optimiser paths

Three ways to forecast the tail



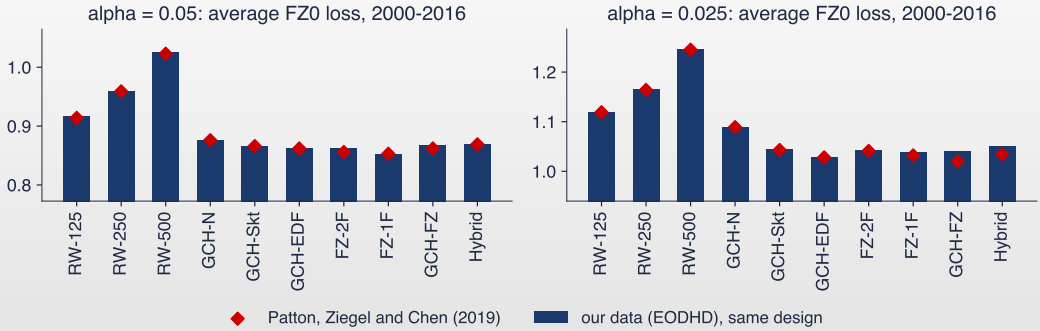
- VaR and ES 5% of the S&P 500 in 2015–2016 (PZC, Figure 5 on our data): RW-125, GARCH-EDF and GAS-1F

ATS_ch9_pzc

Interpreting the three forecasts

- The rolling window moves in steps as extreme days enter and leave the window, months after the shock
- GARCH-EDF moves every day with squared returns; GAS-1F moves only on hit days and then decays smoothly
- August 2015: GARCH reacts most, GAS-1F jumps less but stays; in 2008 the GAS-1F ES 5% reached -9.9%
- Which reaction is right is decided by the FZ0 loss, not by the look of the path

Replication: average out-of-sample FZ0 loss



⊡ Bars: our data and code, same design; diamonds: the values printed in PZC, Table 8 ($\alpha = 5\%$) and Table S5 ($\alpha = 2.5\%$)

ATS_ch9_pzc

Replication in numbers

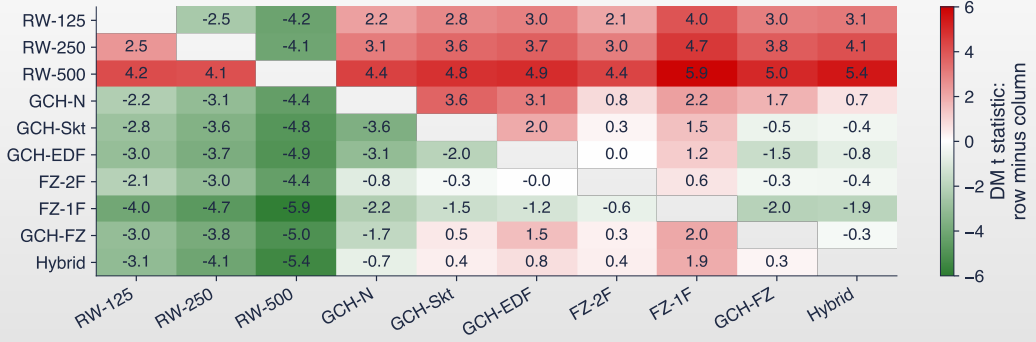
Model	$\alpha = 5\%$		$\alpha = 2.5\%$		GoF p, $\alpha = 5\%$	
	ours	PZC	ours	PZC	VaR	ES
RW-125	0.916	0.914	1.120	1.119	0.004	0.005
RW-250	0.959	0.959	1.165	1.164	$<$0.001	0.011
RW-500	1.025	1.023	1.247	1.245	$<$0.001	$<$0.001
GCH-N	0.876	0.876	1.089	1.089	0.018	$<$0.001
GCH-Skt	0.865	0.866	1.044	1.043	0.001	0.001
GCH-EDF	0.862	0.862	1.028	1.028	0.001	0.007
FZ-2F	0.862	0.856	1.044	1.041	$<$0.001	0.003
FZ-1F	0.853	0.853	1.038	1.032	0.001	0.037
GCH-FZ	0.867	0.862	1.040	1.020	0.005	0.016
Hybrid	0.869	0.869	1.051	1.034	$<$0.001	$<$0.001

- Largest gap: 0.006 at 5%, 0.020 at 2.5%; BIC picks ARMA(0, 0) for the mean (PZC: ARMA(1,1)); GARCH $\alpha = 0.052$, $\beta = 0.942$

Interpreting the replication

- The ranking of the paper is reproduced: GAS-1F lowest at 5% (0.853, paper 0.853), RW-500 worst, GARCH-N behind the other GARCH models
- GAS-1F estimates: $\beta = 0.995$, $\gamma = -0.0069$ (PZC, Table 7: 0.990 and -0.010); Hybrid $\delta = 0.017$ (0.018)
- Not reproduced: the goodness-of-fit tests; PZC find GAS-1F passing (p 0.242 and 0.313), on our data it is rejected (0.001; 0.037)
- Average losses are robust to small data differences; tests on a handful of tail days are not, and the covariance choice of the regression matters

Diebold–Mariano tests on FZ0 losses



- S&P 500, 2000–2016, $\alpha = 5\%$: DM statistic of row minus column, Newey–West variance; red: the row model is worse (PZC, Table 9)

ATS_ch9_pzc

Interpreting the DM matrix

- Column GAS-1F, ours against PZC: RW-125 4.02 (3.91), RW-500 5.87 (5.48), GARCH-N 2.22 (1.99), GARCH-EDF 1.19 (1.20)
- The worst models are easily separated; the best few are not (GARCH-Skt, GARCH-EDF, FZ-2F against GAS-1F below 1.96)
- One difference: GARCH-FZ against GAS-1F, 2.02 here and 1.27 in the paper; the scale normalisation of GARCH-FZ makes its estimates fragile
- Ninety pairwise tests need a multiple-testing answer: the MCS (Section 6)

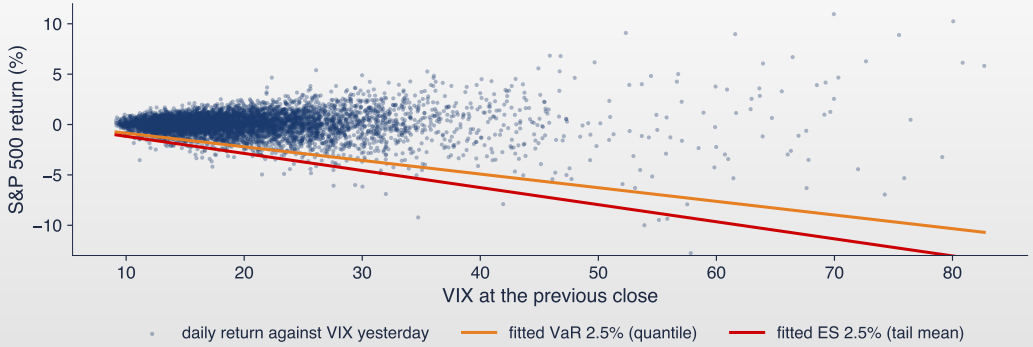
Joint (VaR, ES) regression

- Dimitriadis and Bayer (2019): both functionals are linear in the covariates

$$q_\alpha(Y_t \mid x_t) = x_t'\beta, \qquad -\mathrm{ES}_\alpha(Y_t \mid x_t) = x_t'\gamma$$

  - x_t: covariates known at $t-1$; β, γ: the effects on the quantile and on the tail mean; estimated jointly by minimising a Fissler–Ziegel loss
  - M-estimator: consistent and asymptotically normal for any member of the class; the choice of (G_1, G_2) affects only efficiency
- Uses
  - which variables move the tail and by how much: the ES analogue of quantile regression
  - ES backtests: regress realised returns on the ES forecast and test intercept 0, slope 1 Bayer and Dimitriadis (2022) (MFM, Chapter 8)
- Application: S&P 500 returns 2000–2026 on the VIX of the previous close, $\alpha = 2.5\%$, FZ0 loss, moving-block bootstrap standard errors (blocks of 20 days)

The tail of the S&P 500 against the VIX



⊡ Daily returns against the VIX of the previous close, $T = 6\,714$; lines: fitted 2.5% quantile and tail mean of the joint regression

ATS_ch9_esreg

Interpreting the joint regression

- Quantile: $0.499 - 0.135\,\mathrm{VIX}$ (s.e. 0.150; 0.009); tail mean: $0.524 - 0.170\,\mathrm{VIX}$ (s.e. 0.245; 0.015)
- One VIX point lowers the 2.5% quantile by 0.14 and the tail mean by 0.17 percentage points: the tail fans out 1.25 times faster in ES
- The quantile part is close to quantile regression (0.560, -0.139); in-sample hit rate 2.50%; the PZC tests do not reject ($p = 0.978$; 0.744)
- The VIX is a market forecast of volatility: the regression says that the option market prices the left tail almost linearly

Recap: Semiparametric (VaR, ES) models

- FZ0 minimisation estimates the tail pair without a distribution
- GAS forcing variables are identification functions: the model corrects itself only when the tail speaks
- The PZC ranking replicates; their goodness-of-fit results do not; joint regression links the tail to covariates

Recap: counting hits

- Hits $I_t = 1\{y_t < v_t\}$; a correct VaR makes I_t i.i.d. Bernoulli(α) (MFM, Chapter 8)
  - Kupiec (1995): unconditional coverage, likelihood-ratio statistic LR $\sim \chi^2_1$ (is the hit rate α?)
  - Christoffersen (1998): independence of consecutive hits, and conditional coverage (both together), χ^2_2
  - Basel traffic light Basel Committee (1996): green up to 4 hits of VaR 1% in 250 days, yellow 5–9, red from 10
- What counting misses
  - hits that depend on information other than the last hit (DQ)
  - the size of tail losses (ES backtests); the spacing of hits (duration tests)
  - the effect of estimated parameters on the null distribution

Conditional calibration

- A forecast x_t of T is **conditionally calibrated** if its identification function V has conditional mean zero Nolde and Ziegel (2017)

$$\mathbb{E}[V(x_t, Y_t) \mid \mathcal{F}_{t-1}] = 0 \;\Rightarrow\; \mathbb{E}[h_{t-1} V(x_t, Y_t)] = 0$$

  - h_{t-1}: any instrument known at $t-1$ (constant, lagged hits, the forecast itself); a moment test (Wald, χ^2)
- PZC, eq. (40)–(41): standardised generalised residuals

$$\lambda^s_{v,t} = 1\{y_t \le v_t\} - \alpha, \qquad \lambda^s_{e,t} = \frac{1\{y_t \le v_t\}\, y_t}{\alpha e_t} - 1$$

  - regress each on (1, its own first lag, v_t or e_t) and test that all three coefficients are zero; we use a White (HC0) covariance
  - DQ is the special case for VaR; the ES regression tests ES only jointly with VaR, as elicitability predicts
- Power comes from the instruments: lagged hits detect clustering, the forecast level detects scale errors

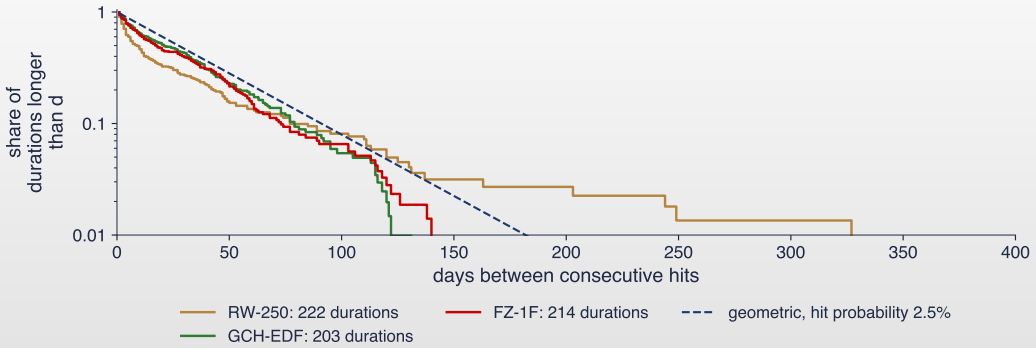
Duration-based backtests

- Christoffersen and Pelletier (2004): under a correct VaR the durations d_i (days between consecutive hits) are geometric, memoryless, with mean $1/\alpha$
  - alternative: Weibull hazard $\lambda(d) = a^b b\, d^{b-1}$, the probability of a hit after d quiet days; $a > 0$: scale, $b > 0$: shape
  - $b = 1$: constant hazard (no memory, as under H_0); $b < 1$: decreasing hazard, i.e. hits cluster
- Likelihood with censoring: the first and last spells are incomplete and enter through the survival function $S(d) = e^{-(ad)^b}$

$$\mathrm{LR} = 2\big[\ln L(\hat{a}, \hat{b}) - \ln L(\tilde{a}, 1)\big] \to \chi^2_1$$

  - $(\hat{a}, \hat{b})$: unrestricted estimates; $\tilde{a}$: estimate under $b = 1$; small-sample p-values by simulation are advisable
- Durations see clustering at any distance, not only on consecutive days as Christoffersen's test does

Durations between VaR hits



⊡ S&P 500, VaR 2.5%, out of sample 2000–2026; empirical survival of the durations between hits (log scale) against the geometric law of a correct model

ATS_ch9_backtests

Interpreting the durations

- RW-250: 40% of the durations are at most 5 days (geometric: 12%); Weibull $b = 0.67$, $p < 0.001$: strong clustering
- GARCH-EDF: $b = 0.92$, $p = 0.107$; GAS-1F: $b = 0.88$, $p = 0.019$: dynamic models remove most of the clustering
- The long tail of RW-250 (calm years without hits) is the other face of clustering: the window remembers old crises
- All models have too many hits overall (Section 5 table): parameters fixed in 1999 are too optimistic after 2000

Backtesting ES: what each test needs

Test	Statistic	Inputs
McNeil and Frey (2000)	mean of $(y_t - e_t)/\sigma_t$ on hit days, bootstrap	v_t, e_t, σ_t
Acerbi and Székely (2014) Z_2	$1 - \frac{1}{T\alpha}\sum_t \frac{y_t I_t}{e_t}$	v_t, e_t and the forecast distribution (simulated p-value)
Du and Escanciano (2017)	cumulative violations $H_t = \frac{1}{\alpha}(\alpha - u_t)\mathbf{1}\{u_t \le \alpha\}$	the PIT $u_t = F_{t\|t-\mathbf{1}}(y_t)$
Kratz, Lok and McNeil (2018)	multinomial test of several VaR levels	the PIT, or VaR at several levels
Patton, Ziegel and Chen (2019), Bayer and Dimitriadis (2022)	regression on (v_t, e_t)	v_t, e_t

- σ_t: forecast volatility; I_t: hit indicator; $u_t = F_{t|t-\mathbf{1}}(y_t)$: probability integral transform (PIT), uniform on $[0, 1]$ under a correct forecast distribution
- Only the regression tests use nothing beyond the pair (VaR, ES): the others are tests of a distribution, which ES alone does not supply

Estimation risk in backtests

- Backtests assume the VaR is known; in practice $\hat{v}_t = v_t(\hat{\theta}_R)$, estimated on R days and tested on the next P days Escanciano and Olmo (2010)

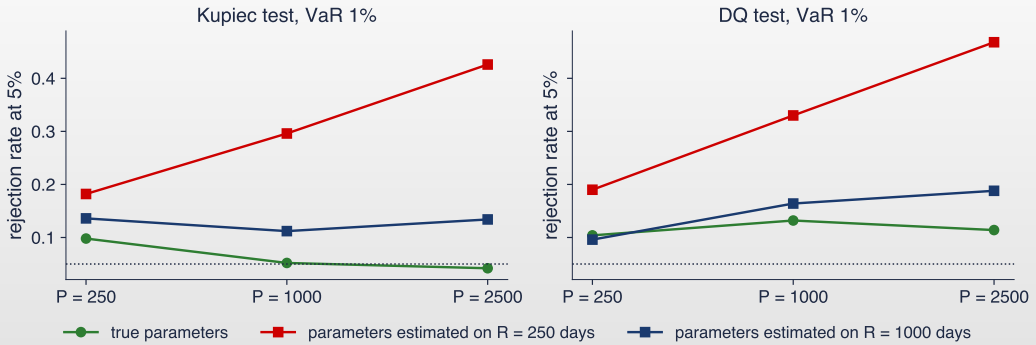
$$\frac{1}{\sqrt{P}}\sum_t(\hat{I}_t - \alpha) = \frac{1}{\sqrt{P}}\sum_t(I_t - \alpha) + \underbrace{\sqrt{P/R}\cdot\mathbb{E}[f_t(v_t)\nabla v_t']\,\sqrt{R}(\hat{\theta}_R - \theta)}_{\text{estimation term}}$$

  - $\hat{I}_t$, I_t: hits of the estimated and of the true VaR; $f_t(v_t)$: conditional density at the VaR; ∇v_t: gradient of the VaR in θ

- The estimation term vanishes only if $P/R \to 0$; with a fixed window it grows with the test sample
  - the Kupiec variance $\alpha(1-\alpha)$ is then too small and the test over-rejects a correct model
  - fixes: the corrected variance of Escanciano and Olmo, a subsampling or bootstrap of the whole estimate-and-test procedure

- A Monte Carlo shows the size of the problem for the windows banks actually use

Size of backtests with estimated parameters



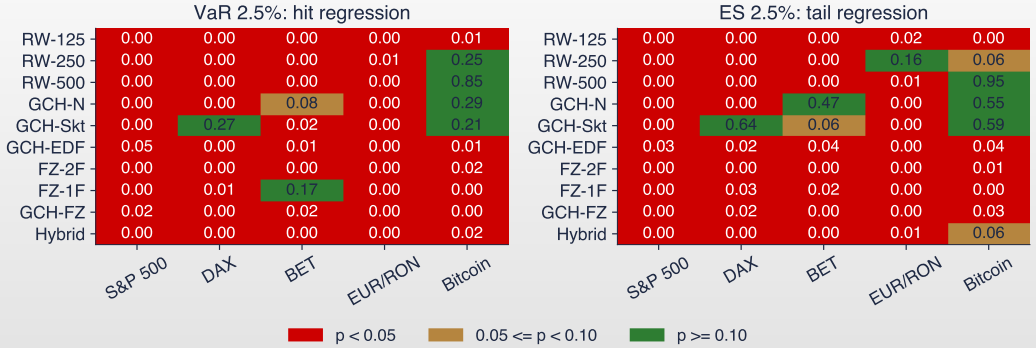
- GARCH(1,1) data ($\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$, Normal), the correct model estimated by QML on R days, VaR 1% on the next P days; 500 replications; nominal size 5%

ATS_ch9_backtests

Interpreting the Monte Carlo

- Known parameters: Kupiec rejects 10%, 5% and 4% for $P = 250$, 1000, 2500: close to 5% except with few hits
- Estimated on $R = 250$: 18%, 30%, 43%: a correct model is rejected more often the longer we test
- With $R = 1000$ the distortion shrinks (13% at $P = 2500$); the hit rate across replications has s.d. 0.99 pp against 0.29 pp
- DQ is oversized even with known parameters (10%): rare hits make the χ^2_6 approximation poor

Calibration tests across assets



- PZC goodness-of-fit regressions for VaR and ES 2.5%, ten models, parameters estimated on the first ten years (five for Bitcoin) and kept fixed to 18 September 2026

ATS_ch9_backtests

Interpreting the calibration tests

- Only 4 of 50 asset–model pairs pass both regressions at 10%: over 10–26 years with fixed parameters almost every model is miscalibrated somewhere
- S&P 500 hit rates at 2.5%: GAS-1F 3.2%, GARCH-EDF 3.0%, RW-250 3.3%: the level drifts after the estimation decade
- EUR/RON: estimated on 2005–2015 (crisis and float), the GARCH models over-predict risk afterwards (GARCH-EDF 0.6% hits): miscalibration in the safe direction is rejected too
- Bitcoin: short history and noisy tail, low power; the simple models pass (RW-500 2.5%)
- A backtest answers "is it calibrated?"; whether to prefer one failing model to another is a comparison question

Recap: Backtesting

- Backtests are moment tests of the identification function with chosen instruments
- Durations catch clustering at any distance; ES tests need more than ES, except the regression tests
- Estimated parameters inflate the size; long, fixed-parameter evaluations reject almost everything

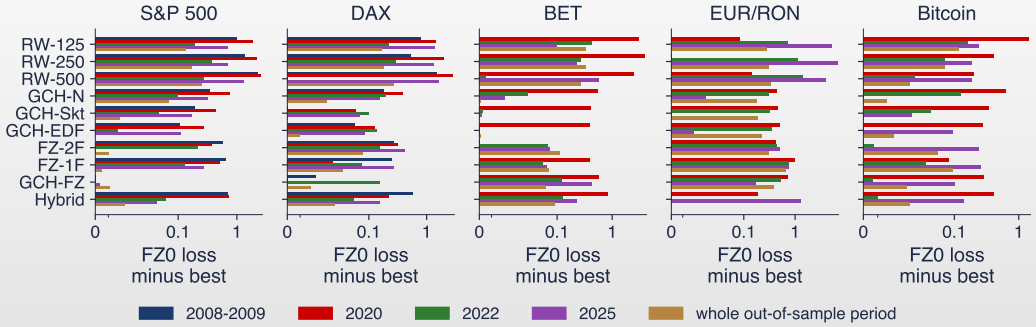
From backtests to comparative backtests

- Nolde and Ziegel (2017): a traditional backtest tests H_0: "the bank's model is calibrated"; a comparative backtest tests the bank's model against a standard one with a consistent score
  - H_0^-: internal at least as good as standard; H_0^+: at most as good; green if H_0^+ is rejected, red if H_0^- is rejected, yellow otherwise
- Information sets matter Holzmann and Eulert (2014): a calibrated forecast based on more information has a lower expected consistent score
  - so a ranking by FZ0 rewards both calibration and information, which a backtest cannot do
- Tools from Chapter 1: DM with HAC variance Diebold and Mariano (1995), Giacomini–White for estimated forecasting methods Giacomini and White (2006), the MCS Hansen, Lunde and Nason (2011)

Design of the comparison

- Ten PZC models, $\alpha = 2.5\%$, FZ0 losses out of sample: S&P 500 and DAX from 2000, BET from 2010, EUR/RON from July 2015, Bitcoin from 2020
- 90% MCS with the $T_{\max}$ statistic, moving-block bootstrap (blocks of 10 days over the whole period, 5 days in a stress period), 1000 replications
- The four stress periods of the data slide: 2008, 2020, 2022 and 2025
  - S&P 500 in 2020: 93 days, of which 9 GAS-1F hits; in 2025: 83 days, 5 hits
- A stress window holds a few tail days: expect wide MCS sets and unstable winners

Losses in stress periods



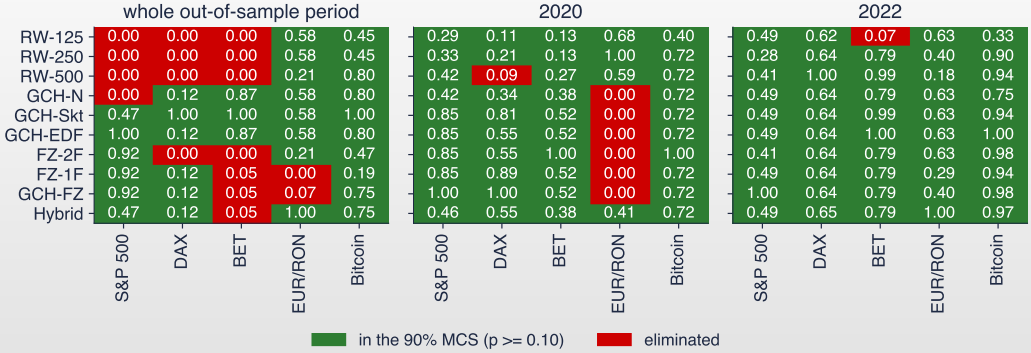
- Average FZ0 loss of each model minus that of the best model in the same period (symmetric log scale); bars missing when the period is not out of sample

ATS_ch9_comparison

Interpreting the stress-period losses

- Winners over the whole period: S&P 500 GCH-EDF, DAX GCH-Skt, BET GCH-Skt, EUR/RON Hybrid, Bitcoin GCH-Skt
- In 2020 the winner changes: S&P 500 GCH-FZ, BET FZ-2F, Bitcoin FZ-2F; rolling windows lose most in every crash
- Crashes separate models by an order of magnitude more than calm years: an average over the whole period is dominated by a few weeks
- No model wins every crisis; the semiparametric models are competitive but not dominant out of sample after 2016

Model confidence sets



- MCS p-values (FZ0 loss, $\alpha = 2.5\%$); green: in the 90% MCS

ATS_ch9_comparison

Interpreting the MCS

- Whole period, MCS sizes: S&P 500 6, DAX 6, BET 3, EUR/RON 8, Bitcoin 10 of 10
- Rolling windows are eliminated on the long equity samples; in 2020 and 2022 almost everything survives (2020: 10 on the S&P 500; 2022: 10)
- EUR/RON in 2020 is the exception: only 4 models survive, the rolling windows and the Hybrid
- A large MCS is a result, not a failure: the crisis windows are too short to separate tail forecasts

Recap: Comparison

- Comparative backtests put the burden of proof on the model, and reward information
- Rankings change in crises; whole-period averages hide this
- Report the MCS, not the winner

Why the square-root-of-time rule fails (1/2)

- The rule scales the one-day VaR to h days

$$\mathrm{VaR}^{(h)} = \sqrt{h}\,\mathrm{VaR}^{(1)}$$

  - $\mathrm{VaR}^{(h)}$: VaR of the h-day return $\sum_{k=1}^{h} y_{t+k}$; exact for i.i.d. returns from the Normal distribution with zero mean
  - for i.i.d. stable laws with index $\gamma \in (0, 2]$ the factor is $h^{1/\gamma}$ ($\gamma = 2$: the Normal case)
- Fat tails that are not stable: the h-day sum is closer to Normal than the daily return
  - so $\sqrt{h}$ overstates the quantile at long horizons Daníelsson and Zigrand (2006)
- Leverage (GJR) makes multi-day losses more skewed than daily ones: the left tail of the sum is thicker than the scaled daily tail

Why the square-root-of-time rule fails (2/2)

- Volatility clustering: for GARCH(1,1) the variance of the h-day return depends on today's volatility

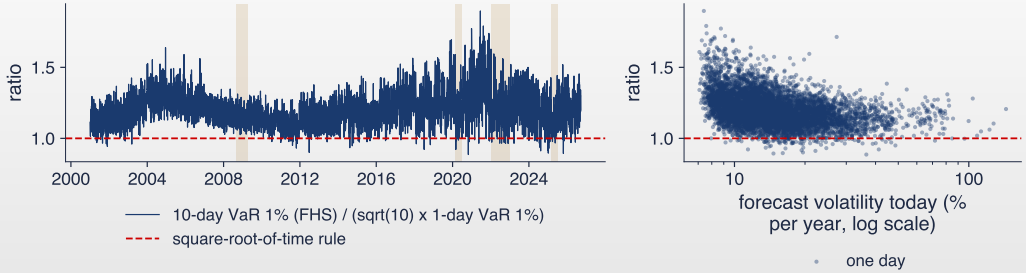
$$\mathrm{Var}_t\Big(\sum_{k=1}^{h} y_{t+k}\Big) = \sum_{k=0}^{h-1} \left[\bar{\sigma}^2 + (\alpha+\beta)^k(\sigma^2_{t+1} - \bar{\sigma}^2)\right]$$

  - $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$: unconditional variance; σ^2_{t+1}: tomorrow's forecast variance; $\alpha+\beta$: persistence
  - in calm times ($\sigma^2_{t+1} < \bar{\sigma}^2$) $\sqrt{h}$ understates risk, in crises it overstates it: mean reversion of volatility
- Remedies: simulate the h-day distribution (FHS), or forecast the h-day quantile directly (quantile regression on h-day returns)

Multi-day forecasts and their backtests

- FHS: GJR-GARCH(1,1) by QML Bollerslev (1986), Glosten, Jagannathan and Runkle (1993)
  - rolling 2000-day window, re-estimated every 250 days
  - 2000 paths of 10 days with bootstrapped standardised residuals
- Backtesting 10-day VaR
  - daily 10-day forecasts overlap: hits are MA(9) even under H_0; counting them as independent inflates the size (Chapter 0, overlapping observations)
  - options: non-overlapping windows (every 10th day, few hits), or HAC variance on overlapping hits
- Basel uses 10-day horizons and liquidity-adjusted horizons for ES (MAR33): the horizon problem is regulatory, not academic

10-day VaR 1% against the square-root-of-time rule



- S&P 500, 2001–2026: ratio of the 10-day VaR 1% by FHS to $\sqrt{10}$ times the 1-day VaR 1% of the same model; right: the ratio against today's forecast volatility

ATS_ch9_horizon

Interpreting the horizon ratio

- Median ratio 1.19, range 0.89 to 1.90: $\sqrt{10}$ scaling usually understates the 10-day VaR 1% of the S&P 500
- Low-volatility days: median 1.26; high-volatility days: 1.14: mean reversion pushes the ratio towards 1 in crises
- Non-overlapping backtest (646 windows, 6.5 expected hits): FHS 6, $\sqrt{10}$ rule 7; Kupiec p 0.85 and 0.83
- So few hits cannot separate the two: horizon errors of this size are invisible to a non-overlapping backtest

Recap: The horizon

- The square-root rule needs i.i.d. Normal returns; clustering, leverage and fat tails break it in opposite directions
- Simulate the horizon or model it directly; backtest with non-overlapping windows or HAC variances
- Multi-day tests have very little power

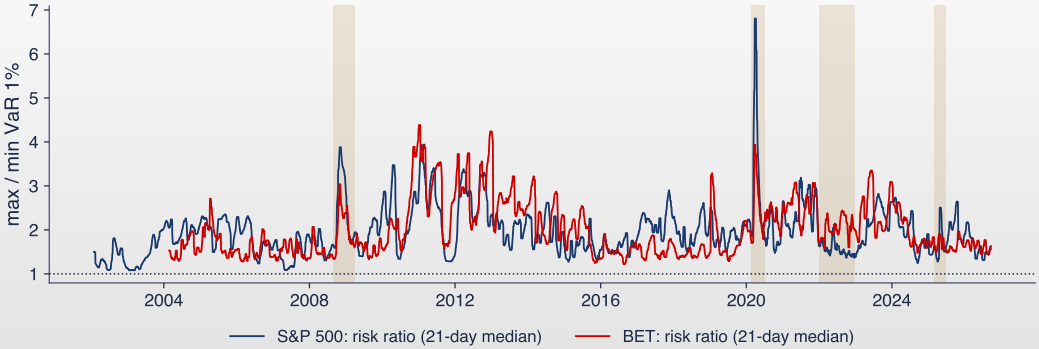
Model risk of risk models

- Danielsson et al. (2016): the **risk ratio** across standard models measures the disagreement a regulator should expect

$$\mathrm{RR}_t = \frac{\max_m \mathrm{VaR}_{m,t}}{\min_m \mathrm{VaR}_{m,t}} \geq 1$$

  - $\mathrm{VaR}_{m,t}$: the forecast of model m for day t; $\mathrm{RR}_t = 1$: all models agree; 2: the most prudent model asks for twice the capital of the least prudent
  - they find the ratio largest exactly in crises, when risk numbers matter most
- Boucher et al. (2014): **risk models-at-risk**: quantify the model risk of a VaR forecast and adjust the forecast for estimation and specification errors
- Kellner and Rösch (2016): ES carries more model risk than VaR at comparable levels: it extrapolates further into the tail
- Our six models, rolling 1000-day windows: historical simulation, Normal with window volatility, EWMA ($\lambda = 0.94$), GARCH-N, GARCH-t, filtered HS

How much do standard models disagree?



- Risk ratio of six VaR 1% forecasts, 21-day rolling median, S&P 500 and BET, 2002–2026 (BET from 2004); shaded: stress periods

ATS_ch9_modelrisk

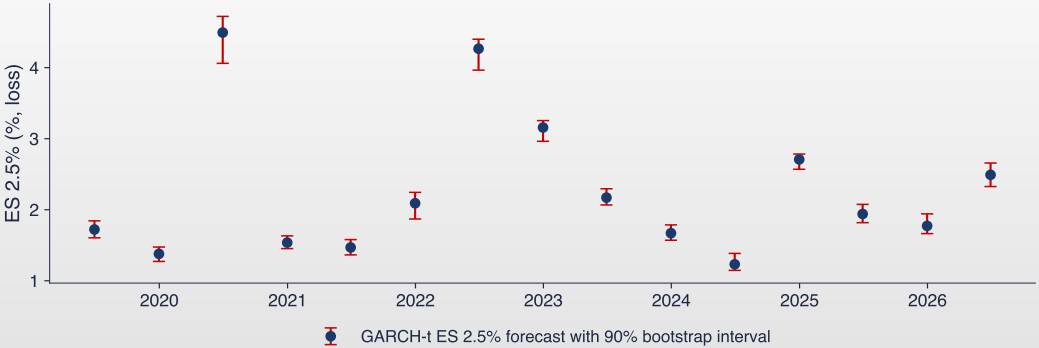
Interpreting the risk ratio

- Median risk ratio 1.9 (S&P 500) and 1.8 (BET); 90th percentile 2.7 and 3.0; maximum 10.5 on 17 March 2020
- Spring 2020: median 2.9 (S&P 500), 2.5 (BET): the slow historical simulation and the fast EWMA diverge most after a jump
- HS is the most conservative model on 73% of S&P 500 days, EWMA the least on 53%
- Hit rates for VaR 1% (S&P 500): HS 1.4%, GARCH-N 2.2%, GARCH-t 1.4%, FHS 1.3%: the disagreement is not noise, the Normal models are wrong

Estimation risk in tail forecasts

- Gao and Song (2008): for GARCH VaR and ES, $\sqrt{T}(\widehat{\mathrm{ES}}_t - \mathrm{ES}_t)$ is asymptotically normal; the variance has a volatility part and a tail-shape part
- Bootstrap Christoffersen and Gonçalves (2005): re-estimate the model on simulated paths and recompute the forecast **conditional on the observed history**
  - algorithm: fit GARCH-t on 2000 days; simulate 150 paths of 2000 days from the fit; re-estimate on each; filter the observed returns with each $\hat{\theta}^*$; ES^* for tomorrow
  - 90% interval: the 5% and 95% quantiles of ES^*
- The interval measures parameter uncertainty only: it is conditional on the model being right, so it is a lower bound on total uncertainty

An ES forecast with its estimation error



- S&P 500, GARCH-t ES 2.5% for the next day at half-year ends 2019–2026, with 90% parametric bootstrap intervals (150 re-estimations each)

ATS_ch9_modelrisk

Interpreting the estimation error

- Interval width relative to the forecast: median 13%, from 8% to 19%
- June 2020: ES 4.49% with interval [4.06; 4.72]: two thousand days of data still leave a band of several tenths of a percent
- Estimation risk is small next to model risk (risk ratios near 2): the choice of model matters more than its precision
- Capital rules that multiply a point forecast ignore both: reporting the interval is the honest minimum

Recap: Model and estimation risk

- Standard models disagree by a factor of about two, more in crises
- Bootstrap intervals for ES are narrow next to that disagreement
- Report both: the forecast, its interval and the spread across models

Extremes of dependent series

- For a stationary series, dependence changes the law of the maximum only through one number, the **extremal index** θ Embrechts, Klüppelberg and Mikosch (1997)

$$\Pr\left(\max_{t\le n} X_t \le u_n\right) \approx F(u_n)^{n\theta}, \qquad \theta \in (0,1]$$

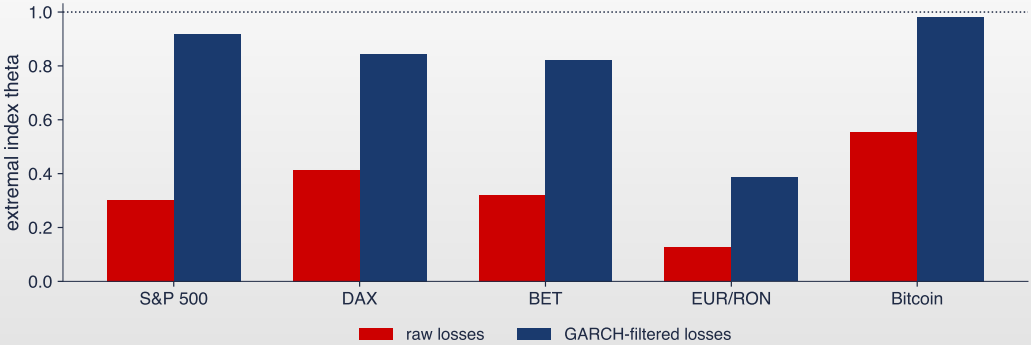
  - X_t: the daily loss; F: its marginal distribution; u_n: a high threshold that grows with the sample size n
  - $1/\theta$ = mean cluster size of exceedances; $\theta = 1$: no clustering of extremes

- Intervals estimator Ferro and Segers (2003), from the gaps T_i between the N exceedances (when $\max T_i > 2$)

$$\hat{\theta} = \min\left(1, \frac{2\left(\sum_i (T_i - 1)\right)^2}{(N-1)\sum_i (T_i - 1)(T_i - 2)}\right)$$

- McNeil and Frey (2000): fit GARCH, apply EVT to the standardised residuals; it works if filtering removes the clustering (θ near 1 after filtering)

Extremal index before and after filtering



- Exceedances of daily losses over their 95% quantile, whole samples; raw losses and GARCH-standardised losses (in-sample ARMA-GARCH parameters)

ATS_ch9_extremes

Interpreting the extremal index

- Raw losses: $\hat{\theta} = 0.30$ (S&P 500), 0.32 (BET), 0.13 (EUR/RON): extremes come in clusters of two to eight days
- After filtering: 0.92, 0.85, 0.98 for the S&P 500, DAX and Bitcoin: GARCH removes most of the clustering, as McNeil and Frey assume
- EUR/RON stays clustered (0.39) and BET partly (0.82): a managed exchange rate and a thin market have dependence that GARCH does not capture
- With fixed in-sample parameters the filter is imperfect after the estimation period: part of the remaining clustering is drift

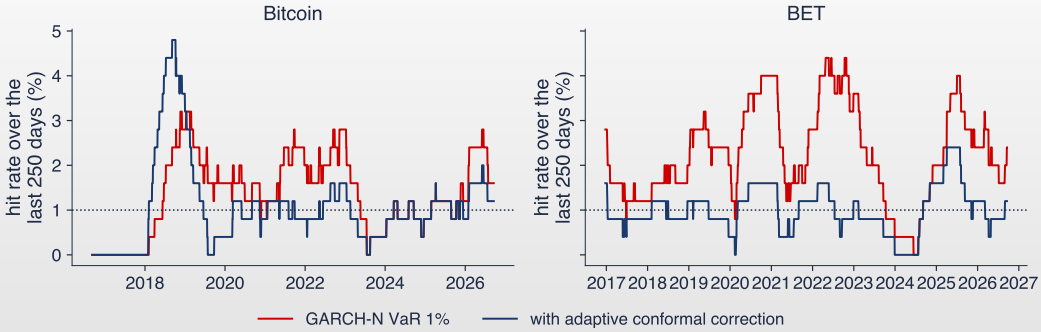
Conformal calibration of VaR

- Split conformal quantiles guarantee coverage under exchangeability (Chapter 13); returns are not exchangeable, so the guarantee fails
- **Adaptive conformal inference** (ACI) Gibbs and Candès (2021): forecast at a working level α_t and correct it after each day

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t), \qquad \mathrm{err}_t = 1\{y_t < q_t(\alpha_t)\}$$

  - $q_t(\alpha_t)$: the model's quantile at level α_t; $\gamma > 0$: step size; after a hit α_{t+1} falls (VaR rises), after a quiet day it rises slightly
  - deterministic guarantee: $\left|T^{-1}\sum_t \mathrm{err}_t - \alpha\right| \le \dfrac{\max(\alpha_1, 1-\alpha_1)+\gamma}{\gamma T}$ for **any** sequence of returns (α_1: the starting level)
  - long-run frequency only: no conditional calibration, no statement about ES
- Further reading: conformal recalibration of extreme tail quantiles under dependence Pele et al. (Conformal_Oracle)

Adaptive conformal correction of VaR 1%



- GARCH-N VaR 1% (rolling 1000-day window, re-estimated every 250 days) from 2016, and its ACI correction with $\gamma = 0.005$; rolling 250-day hit rates

ATS_ch9_conformal

Interpreting the conformal correction

- BET: GARCH-N hits 2.36% of 2 675 days (Kupiec $p < 0.001$); with ACI 1.08% ($p = 0.666$), conditional coverage $p = 0.662$
- Bitcoin: 1.28% to 1.02%; the rolling hit rate is not smoothed: its maximum rises from 3.2% to 4.8% after over-correction
- ACI fixes the frequency, not the dynamics: the adapted level can go below zero (VaR then infinite) after a run of quiet days
- A useful wrapper for a regulator's count; a calibrated model is still needed for DQ and for ES

Recap: Extremes and conformal calibration

- Extremes cluster; filtering removes most clustering for liquid markets, not for managed rates
- ACI guarantees the long-run hit frequency for any data, nothing more
- Both are corrections of a model, not replacements for one

An open question

- Do semiparametric (VaR, ES) models beat location–scale models out of sample once estimation is repeated, and is the advantage concentrated in crises?
  - formal: H_0: equal expected FZ0 loss of GAS-1F and GARCH-EDF with rolling re-estimation, on pre-registered assets, periods and levels
  - falsified by a significant GW statistic in the pre-registered crisis windows, robust across $\alpha = 1\%, 2.5\%, 5\%$
- Why it matters: Basel requires ES models; the PZC evidence comes from fixed parameters and four indices
  - literature to start from: Patton, Ziegel and Chen (2019), Taylor (2019), Nolde and Ziegel (2017), Dimitriadis and Bayer (2019)

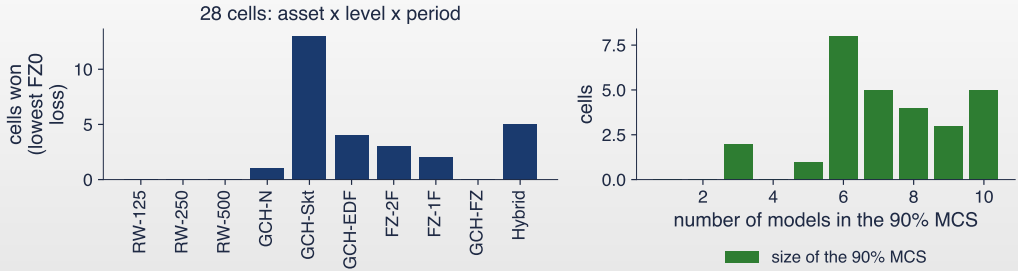
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers since 2017 that compare dynamic ES models out of sample with FZ losses; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Under which data-generating processes should a one-factor GAS model beat GARCH with empirical innovations for ES 2.5%?`
  - **code and replication**: ask for the GAS-1F recursion, then reproduce a published number first (PZC, Table 8: 0.853)
  - **critique**: `Act as a hostile referee: list the ways an FZ0 ranking could be an artefact of the sample, the level or the starting values.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The loss is consistent for the target: no "average ES error", no MSE of VaR forecasts
- The convention is right: VaR 1% is minus the 1% quantile of returns, never "VaR 99%"; signs of v and e are consistent
- Forecasts use only information available at $t-1$; the estimation window and re-estimation scheme are stated
- Rankings are reported with DM or MCS uncertainty and checked on a Murphy diagram

Mini-case: ES model rankings across assets, levels and periods



- 28 cells: five assets × two levels (2.5%, 5%) × up to three periods (whole, before 2020, from 2020); left: the model with the lowest FZ0 loss; right: size of the 90% MCS
- 6 different winners; GARCH-Skt wins 13 cells, GAS-1F 2; MCS sizes from 3 to 10, median 7; GAS-1F is in the MCS in 22 cells: an AI summary that names "the best ES model" is wrong

ATS_ch9_comparison

Project idea

- **ES models for Central and Eastern European markets under rolling re-estimation**: replicate first, then extend
  - replicate: PZC Table 8 and Table S5 for the S&P 500 (GAS-1F 0.853 and 1.032) and the EM design with VaR 1%
  - extend: BET, WIG20, BUX, PX and EUR/RON; rolling re-estimation; joint (VaR, ES) regressions with VIX and the BNR rate; ACI as a wrapper
  - pre-register: samples, levels, models, windows, the stress periods and the MCS design
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- A risk forecast is a point forecast of a functional; judge it with a loss consistent for that functional
- VaR is elicitable, ES only with VaR; FZ0 is the scale-free choice for time series
- CAViaR and GAS-FZ models estimate the tail directly; their published rankings replicate, their published tests do not
- Backtests are moment tests with estimation risk; comparisons need DM, MCS and Murphy diagrams
- Horizon, model and estimation risk are larger than most reported confidence suggests

Self-assessment

Questions

- Why does a non-convex level set rule out a strictly consistent loss for ES?
- Which choice of (G_1, G_2) gives FZ0, and what does zero homogeneity buy?
- Why is the DQ test with a VaR regressor more powerful than Christoffersen's test?
- Why does a fixed estimation window inflate the size of the Kupiec test?
- When does $\sqrt{10}$ scaling overstate the 10-day VaR?

Next:

- Long memory and rough volatility
- ARFIMA, fractional cointegration, rough volatility

Appendix: consistency of GPL quantile scores

- The expected GPL score of a report x under $Y \sim F$

$$\mathbb{E}_F S(x, Y) = \int_{-\infty}^{x} (1-\alpha)(G(x) - G(y))\, dF(y) + \int_{x}^{\infty} \alpha(G(y) - G(x))\, dF(y)$$

- Differentiate (Leibniz; the boundary terms vanish)

$$\frac{d}{dx}\mathbb{E}_F S(x, Y) = G'(x)\big[(1-\alpha)F(x) - \alpha(1-F(x))\big] = G'(x)(F(x) - \alpha)$$

- $G' \geq 0$: the derivative is ≤ 0 for $x < q_\alpha$ and ≥ 0 for $x > q_\alpha$, so q_α minimises; strictly if G is strictly increasing and F has a unique α-quantile
- The same computation with $G(x) = x$ is the first-order condition of quantile regression

Back

Appendix: FZ0, consistency and homogeneity

- Fix e; in v, FZ0 is $-\frac{1}{\alpha e}\big[1\{y \le v\}(v-y) - \alpha v\big]$ plus terms free of v: since $-1/(\alpha e) > 0$, a positive multiple of the pinball loss, minimised at q_α
- At $v = q_\alpha$: $\mathbb{E}\, L = \dfrac{c}{e} + \ln(-e) - 1$ with $c = v - \frac{1}{\alpha}\mathbb{E}[1\{Y \le v\}(v-Y)] = \mathbb{E}[Y \mid Y \le q_\alpha] = -\mathrm{ES}_\alpha$; the derivative $-c/e^2 + 1/e$ vanishes at $e = c$
- Homogeneity: replacing (y, v, e) by (sy, sv, se), $s > 0$, changes FZ0 only by $\ln s$, the same for every forecast: loss differences are scale-free
- Joint minimisation over (v, e) with $e \le v < 0$ therefore returns $(q_\alpha, -\mathrm{ES}_\alpha)$

Back

References (1/4)

- Acerbi, C., & Székely, B. (2014). Backtesting expected shortfall. *Risk*, December 2014 (MSCI working paper). msci.com
- Acerbi, C., & Tasche, D. (2002). On the coherence of expected shortfall. *Journal of Banking & Finance*, 26(7), 1487–1503. doi:10.1016/S0378-4266(02)00283-2
- Artzner, P., Delbaen, F., Eber, J.-M., & Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203–228. doi:10.1111/1467-9965.00068
- Basel Committee on Banking Supervision (2019). *MAR33: Internal models approach, capital requirements calculation*. Basel Framework, Bank for International Settlements. bis.org
- Basel Committee on Banking Supervision (1996). *Supervisory framework for the use of "backtesting in conjunction with the internal models approach to market risk capital requirements*. Bank for International Settlements. bis.org
- Bayer, S., & Dimitriadis, T. (2022). Regression-based expected shortfall backtesting. *Journal of Financial Econometrics*, 20(3), 437–471. doi:10.1093/jjfinec/nbaa013
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- Boucher, C. M., Daníelsson, J., Kouontchou, P. S., & Maillet, B. B. (2014). Risk models-at-risk. *Journal of Banking & Finance*, 44, 72–92. doi:10.1016/j.jbankfin.2014.03.019
- Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862. doi:10.2307/2527341
- Christoffersen, P., & Gonçalves, S. (2005). Estimation risk in financial risk management. *The Journal of Risk*, 7(3), 1–28. doi:10.21314/JOR.2005.112
- Christoffersen, P., & Pelletier, D. (2004). Backtesting value-at-risk: A duration-based approach. *Journal of Financial Econometrics*, 2(1), 84–108. doi:10.1093/jjfinec/nbh004
- Creal, D., Koopman, S. J., & Lucas, A. (2013). Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5), 777–795. doi:10.1002/jae.1279
- Daníelsson, J., & Zigrand, J.-P. (2006). On time-scaling of risk and the square-root-of-time rule. *Journal of Banking & Finance*, 30(10), 2701–2713. doi:10.1016/j.jbankfin.2005.10.002

References (2/4)

- Danielsson, J., James, K. R., Valenzuela, M., & Zer, I. (2016). Model risk of risk models. *Journal of Financial Stability*, 23, 79–91. doi:10.1016/j.jfs.2016.02.002
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- Dimitriadis, T., & Bayer, S. (2019). A joint quantile and expected shortfall regression framework. *Electronic Journal of Statistics*, 13(1), 1823–1871. doi:10.1214/19-EJS1560
- Du, Z., & Escanciano, J. C. (2017). Backtesting expected shortfall: Accounting for tail risk. *Management Science*, 63(4), 940–958. doi:10.1287/mnsc.2015.2342
- Ehm, W., Gneiting, T., Jordan, A., & Krüger, F. (2016). Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society: Series B*, 78(3), 505–562. doi:10.1111/rssb.12154
- Embrechts, P., Klüppelberg, C., & Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer. doi:10.1007/978-3-642-33483-2
- Engle, R. F., & Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381. doi:10.1198/073500104000000370
- Escanciano, J. C., & Olmo, J. (2010). Backtesting parametric value-at-risk with estimation risk. *Journal of Business & Economic Statistics*, 28(1), 36–51. doi:10.1198/jbes.2009.07063
- Ferro, C. A. T., & Segers, J. (2003). Inference for clusters of extreme values. *Journal of the Royal Statistical Society: Series B*, 65(2), 545–556. doi:10.1111/1467-9868.00401
- Fissler, T., Ziegel, J. F., & Gneiting, T. (2016). Expected shortfall is jointly elicitable with value at risk: Implications for backtesting. *Risk*, January 2016, 58–61; arXiv:1507.00244. arxiv.org
- Fissler, T., & Ziegel, J. F. (2016). Higher order elicitability and Osband's principle. *The Annals of Statistics*, 44(4), 1680–1707. doi:10.1214/16-AOS1439
- Gaglianone, W. P., Lima, L. R., Linton, O., & Smith, D. R. (2011). Evaluating value-at-risk models via quantile regression. *Journal of Business & Economic Statistics*, 29(1), 150–160. doi:10.1198/jbes.2010.07318
- Gao, F., & Song, F. (2008). Estimation risk in GARCH VaR and ES estimates. *Econometric Theory*, 24(5), 1404–1424. doi:10.1017/S0266466608080559

References (3/4)

- Giacomini, R., & White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578. doi:10.1111/j.1468-0262.2006.00718.x
- Gibbs, I., & Candès, E. (2021). Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, 34; arXiv:2106.00170. arxiv.org
- Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5), 1779–1801. doi:10.1111/j.1540-6261.1993.tb05128.x
- Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494), 746–762. doi:10.1198/jasa.2011.r10138
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. doi:10.1198/016214506000001437
- Hansen, B. E. (1994). Autoregressive conditional density estimation. *International Economic Review*, 35(3), 705–730. doi:10.2307/2527081
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497. doi:10.3982/ECTA5771
- Holzmann, H., & Eulert, M. (2014). The role of the information set for forecasting – with applications to risk management. *The Annals of Applied Statistics*, 8(1), 595–621. doi:10.1214/13-AOAS709
- Kellner, R., & Rösch, D. (2016). Quantifying market risk with value-at-risk or expected shortfall? Consequences for capital requirements and model risk. *Journal of Economic Dynamics and Control*, 68, 45–63. doi:10.1016/j.jedc.2016.05.002
- Koenker, R., & Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1), 33–50. doi:10.2307/1913643
- Koenker, R. (2005). *Quantile Regression*. Cambridge University Press. doi:10.1017/CBO9780511754098
- Koenker, R., & Xiao, Z. (2006). Quantile autoregression. *Journal of the American Statistical Association*, 101(475), 980–990. doi:10.1198/016214506000000672
- Kratz, M., Lok, Y. H., & McNeil, A. J. (2018). Multinomial VaR backtests: A simple implicit approach to backtesting expected shortfall. *Journal of Banking & Finance*, 88, 393–407. doi:10.1016/j.jbankfin.2018.01.002

References (4/4)

- Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives*, 3(2), 73–84. doi:10.3905/jod.1995.407942
- McNeil, A. J., & Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300. doi:10.1016/S0927-5398(00)00012-8
- Nolde, N., & Ziegel, J. F. (2017). Elicitability and backtesting: Perspectives for banking regulation. *The Annals of Applied Statistics*, 11(4), 1833–1874. doi:10.1214/17-AOAS1041
- Patton, A. J., Ziegel, J. F., & Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and value-at-risk). *Journal of Econometrics*, 211(2), 388–413. doi:10.1016/j.jeconom.2018.10.008
- Pele, D. T., Bolovăneanu, V., Ginavar, A., Lessmann, S., & Härdle, W. K. Conformal recalibration of extreme tail quantiles under temporal dependence. Working paper and replication package (further reading). github.com
- Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., et al. (2022). Forecasting: Theory and practice. *International Journal of Forecasting*, 38(3), 705–871. doi:10.1016/j.ijforecast.2021.11.001
- Taylor, J. W. (2008). Estimating value at risk and expected shortfall using expectiles. *Journal of Financial Econometrics*, 6(2), 231–252. doi:10.1093/jjfinec/nbn001
- Taylor, J. W. (2019). Forecasting value at risk and expected shortfall using a semiparametric approach based on the asymmetric Laplace distribution. *Journal of Business & Economic Statistics*, 37(1), 121–133. doi:10.1080/07350015.2017.1281815
- Weber, S. (2006). Distribution-invariant risk measures, information, and dynamic consistency. *Mathematical Finance*, 16(2), 419–441. doi:10.1111/j.1467-9965.2006.00277.x
- Ziegel, J. F. (2016). Coherence and elicitability. *Mathematical Finance*, 26(4), 901–918. doi:10.1111/mafi.12080